

Efficient Self-Adaptive Pseudo-Resistor with Rapid Settling and High Linearity for Neurorecording Front-End Circuits

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Abstract—In this paper, we present a novel self-adaptive pseudo-resistor (A-PR) designed to enhance the performance of neurorecording front-end circuits in terms of settling time, linearity, and tunability. We validate the effectiveness of the proposed A-PR through the implementation of a capacitively-coupled instrumentation amplifier (CCIA) recording front-end using TSMC 40-nm process technology. The results demonstrate that the A-PR enables continuous recording with minimal interruptions, enhancing the system's robustness and enabling more reliable acquisition of neural signals. Notably, the A-PR achieves a significant reduction in settling time, reaching the millisecond level—1000 times faster than conventional pseudo-resistors—while also exhibiting wide linear characteristics and easy tunability.

Index Terms—Neural recording, Pseudo-resistor, Fast settling, Stimulation, Tunable amplifier.

I. INTRODUCTION

RECENT advancements in biomedical front-end circuits have played a crucial role in recording electrical activity from the human body [1]–[3]. These circuits amplify small-amplitude and low-frequency neural signals while minimizing noise interference. One challenge in neural amplifier designs is eliminating the dc electrode offset (EDO) [4]–[6], which can cause saturation in the amplifier due to the wide range of intrinsic DC potentials from neurons [7]. The capacitively-coupled instrument amplifier (CCIA) is widely used to address the EDO issue. However, it often requires a large input capacitor to achieve a low high-pass (HP) corner frequency (<0.5 Hz) and minimize input mismatch.

To achieve a low HP corner frequency, pseudo-resistors (PRs) are frequently employed in CCIA designs due to their negligible silicon area [8]–[10]. However, the variation and voltage dependence of PRs often necessitate setting the HP corner frequency to a value significantly lower than 0.5 Hz, resulting in a large time constant and prolonged settling time for the amplifier's operating points. As a result, any potential saturation of the amplifier, such as electrode offset (EDO), motion artifact, or stimulation artifact, can result in an inconsistent output signal [11]–[15].

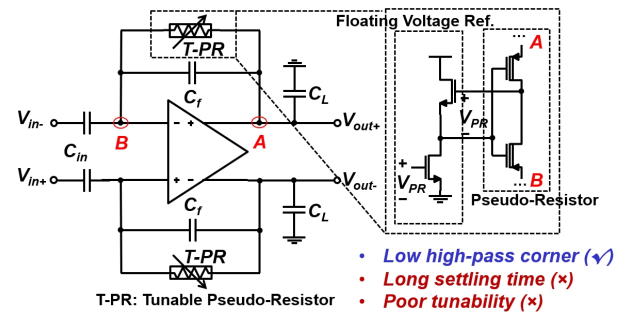


Fig. 1. Schematic of an AC-coupled neural amplifier with its tunable pseudo-resistors.

Fig. 1 illustrates a typical CCIA topology incorporating tunable pseudo-resistors (T-PRs) in the feedback loop [16]–[19]. The static operating point of node A is set by the integrated common mode feedback loop in the low-noise amplifier, while node B's operating point is determined by the DC feedback via the T-PRs. Therefore, achieving a rapid settling of node B is crucial for long-time neural signal recording. Although pseudo-resistors exhibit faster settling behavior compared to large resistors since their resistance is inversely proportional to the applied voltage, the settling time can still reach tens of seconds when the DC voltage across the PRs is low (in the tens of millivolts range). Furthermore, the voltage-dependent resistance of the PRs may introduce non-linearities into the system that must be carefully considered [20]–[22]. To compensate for process variation and enable impedance tunability, a precise and constant V_{PR} is generated by the floating voltage reference, as demonstrated in [8], [10], [23].

In this paper, we present a solution to the issue of long settling time in PRs through a novel bias scheme. The key idea is to build an efficient self-adaptive PR (A-PR) that dynamically adjusts its resistance based on the voltage of the unsettled terminal. Once the terminal reaches a settled state, the PRs exhibit wide linear and symmetrical resistance behavior. To validate the efficacy of the proposed approach, a

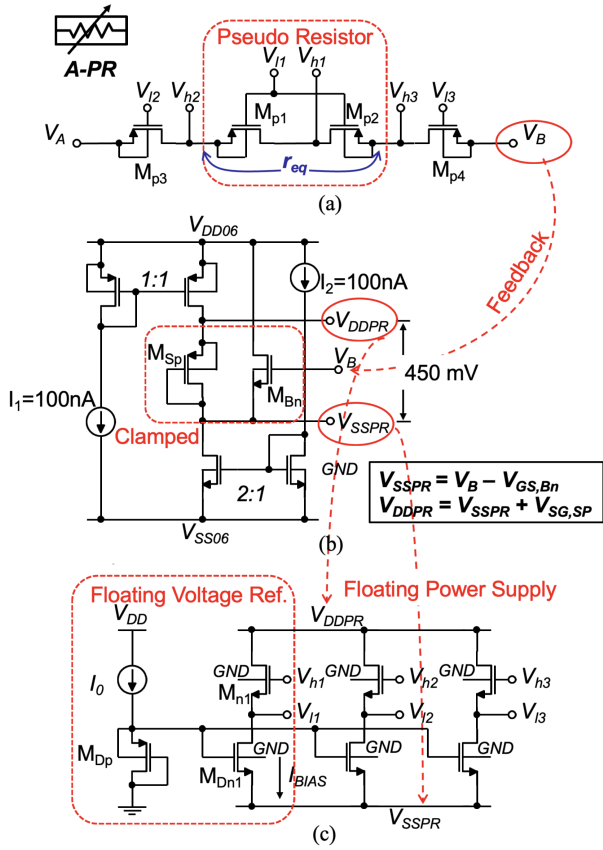


Fig. 2. Schematic of the proposed self-adaptive pseudo-resistor (A-PR). (a) Structure of the tunable pseudo-resistor [10]; (b) Schematic of the adaptive floating power supply (FPS); (c) Floating voltage reference for pseudo-resistor biasing.

CCIA recording front-end equipped with the proposed A-PRs was implemented using TSMC 40-nm process.

This paper is organized as follows: Section II presents the circuit implementation for the A-PR and CCIA. Section III delves into the simulation and measurement outcomes, while Section IV offers concluding remarks.

II. CIRCUIT DESIGN

A. Self-Adaptive Pseudo-Resistor

Various implementations of PRs have been reported in the literature [24]–[27], focusing on improving characteristics such as tunability, linearity, maximum tolerated voltage, and robustness to process variations. However, these existing approaches have not simultaneously achieved a satisfactory combination of high linearity, fast settling time, and easy tunability. In this work, we address these limitations by proposing a self-adaptive PR (A-PR) implementation, thereby bridging the existing gap in PR design.

In Fig. 2(a), the schematic of the tunable PR we considered is depicted, which has demonstrated favorable linearity within a suitable voltage drop range for amplified biosignals, as reported in [10]. Fig. 2(c) shows the bias circuit that provides a constant gate-drain voltage for the PR structure depicted in

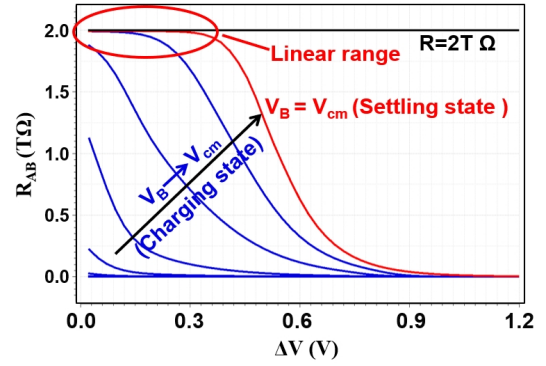


Fig. 3. Resistance of the proposed adaptive pseudo-resistor (A-PR) at varying feedback voltages V_B .

Fig. 2(a). Additionally, the schematic in Fig. 2(b) generates a pair of floating power supplies (V_{DDPR} and V_{SSPR}) to support the biasing circuit in Fig. 2(c). It is worth noting that V_{DD} is set to 1.2 V, while V_{DD06} and V_{SS06} are ± 0.6 V, respectively.

During operation, there are two distinct states: the charging state and the settling state. In the charging state, assuming that V_A is connected to the source input, then V_B is the designated waiting node for charging. The two floating power supplies are then locked to the increasing voltage V_B as it serves as the feedback control for the adaptive floating power supply (FPS) circuit.

To establish the voltage levels, we have $V_{SSPR} = V_B - V_{GS_{Bn}}$ and $V_{DDPR} = V_{SSPR} + V_{SG_{Sp}}$. The biasing of the clamped transistors M_{Sp} and M_{Bn} is achieved using two pairs of current mirrors, resulting in $I_{SD_{Sp}} = I_1 = 100$ nA and $I_{DS_{Bn}} = 2 \times I_2 - I_{SD_{Sp}} = 100$ nA. Consequently, constant values of $V_{GS_{Bn}}$ and $V_{SG_{Sp}}$ are generated, and the voltages V_{DDPR} and V_{SSPR} are solely influenced by the feedback voltage V_B , to which they are directly proportional.

In this design, $V_{DDPR} - V_{SSPR} = V_{SG_{Sp}}$ is set to 450 mV to ensure sufficient headroom for all transistors in the FPS circuit and provide adequate power for the floating voltage reference. The objective is to leverage the non-linearity of the PR, resulting in a significant reduction in its resistance during the charging state, which is inversely proportional to the voltage difference $V_A - V_B$. As V_{DDPR} and V_{SSPR} dynamically increase with the charging of V_B , the generated floating reference voltages V_h and V_l vary from negative to positive, leading to a substantial reduction in the equivalent resistance of the PR during the charging state. Moreover, any mismatch or process variation during this state has minimal impact on the final settling state, as the precision biasing is solely determined by the settling state.

In the settling state, once V_B has reached a stable value, the operation of the PR follows a similar principle as described in the designs presented in [8], [10]. The biasing current I_0 is converted into a source -ground voltage using transistor M_{Dp} . This voltage obtained is subsequently employed to bias the gate of transistor M_{Dn1} , generating the bias current for

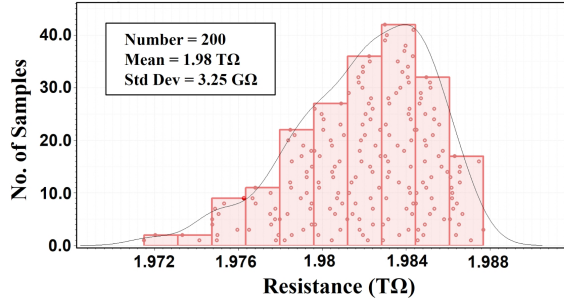


Fig. 4. Simulation results of 200 Monte Carlo runs on the equivalent resistance obtained from the A-PR at 300K.

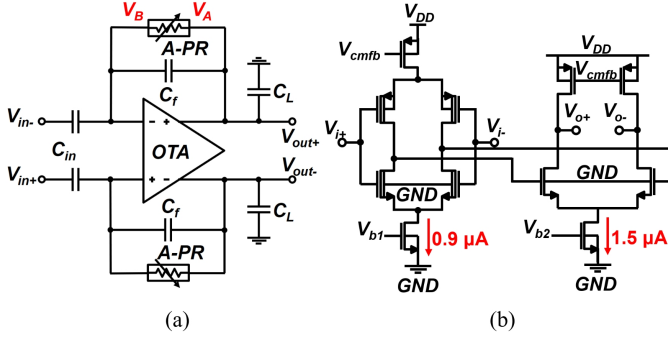


Fig. 5. (a) Schematic of the CCIA recording front-end. (b) Schematic of the OTA.

transistor M_{n1} . The gate-source voltage from transistor M_{n1} is employed to bias the PRs depicted in Fig. 2(a).

When the transistors M_{Dp} , M_{Dn1} , and M_{n1} are equally sized, the PR enclosed within the red dotted box can be represented by a simplified equivalent resistance $r_{eq} |_{V_{h2}=V_{h3}} = \frac{2\phi_t}{I_0}$. In this expression, ϕ_t represents the thermal voltage, and a PTAT (Proportional To Absolute Temperature) current source can compensate for temperature variations. Consequently, the final resistance remains resilient against variations arising from both process corner effects and random process variations.

Fig. 3 illustrates the Resistance-Voltage characteristics of the A-PR for various feedback voltages V_B . Due to the full symmetry of the PR structure, only voltage drops greater than zero are displayed in the figure. As V_B charges towards the settling voltage V_{cm} , the resistance gradually increases until reaching the final settling state, as depicted by the red curve. In the settling state, a linear range spanning from -0.3 V to 0.3 V is achieved, which satisfies the requirements of most biomedical front-end designs.

Fig. 4 demonstrates the impact of transistor mismatch on the resistance of the A-PR. A Monte Carlo simulation comprising 200 runs was conducted to assess this effect. The simulation results reveal that the resistance exhibits a mean value of 1.98 TΩ with a standard deviation of 3.25 GΩ, which is less than 0.2%. This indicates that the A-PR exhibits minimal sensitivity to transistor mismatch, ensuring robust performance in the face of process variations.

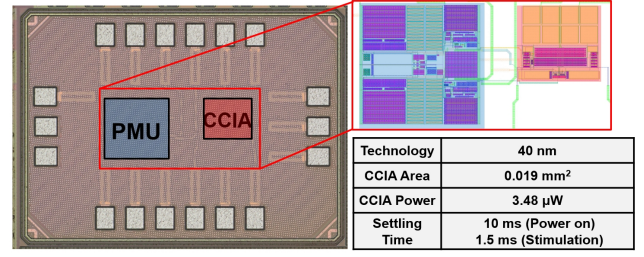


Fig. 6. Micrograph and layout of the chip, featuring the power management unit (PMU) and the capacitively-coupled instrumentation amplifier (CCIA).

B. Biopotential Amplifier

To assess the effectiveness of the A-PR, we integrated it into the feedback network of a differential OTA to serve as the front-end stage of a biopotential recording system, as shown in Fig. 5(a). The schematic of the designed OTA is shown in Fig. 5(b). The front-end comprises a negative feedback capacitor to achieve a precise closed-loop gain of 37.4 dB ($C_{in}/C_f = 19.2$ pF / 257.2 fF) with a high-pass corner (typical 0.5 Hz) set by the A-PR and the feedback capacitor. The differential input transistors are biased in the subthreshold region ($g_m/I_D \approx 27$), and thick-oxide transistors are incorporated to prevent gate leakage. Note that the common-mode feedback circuit is not depicted in the figure.

To eliminate the impact of small signals on linearity once the circuit has settled, we connect the V_B node to the input of the OTA. The supply voltage V_{DD} of the amplifier is 1.2 V, and the output common-mode voltage is fixed at 0.6 V. Therefore, during the settling state of the A-PR, the voltage $V_A = V_B = V_{cm} = 0.6$ V. To expedite the settling time, $V_{GS_{Bn}}$ is specifically set to 0.6 V, causing the voltage V_{SSPR} to be 0 V in the settling state. This also guarantees the matching between M_{Dp} and M_{Dn1} . Moreover, the bulk effect of transistor M_{Bn} enables $V_{GS_{Bn}}$ to be less than 0.6 V during the charging state, allowing the NMOS current mirror to function correctly at the initial stage of the charging process. Furthermore, by lowering V_{SSPR} in the settling state through an increase in the biasing current I_2 , the resistance decreases, allowing for adjustment of the HP corner frequency to suit different applications.

III. SIMULATION AND MEASUREMENT RESULTS

The complete biomedical front-end system was realized by integrating the biopotential amplifier, incorporating the proposed adaptive pseudo-resistor (A-PR), with a power management unit (PMU). Fabricated in TSMC 40-nm process, the amplifier consumes a power of 3.48 μW and occupies a silicon area of 0.019 mm². Notably, the A-PR and its associated biasing circuits only account for approximately 10% of the total power consumption and occupies less than 6% of the chip area. These percentages can be further reduced when multiplexing with more channels. Fig. 6 presents a die micrograph of the chip, showcasing its main blocks, along with a brief summary of its features..

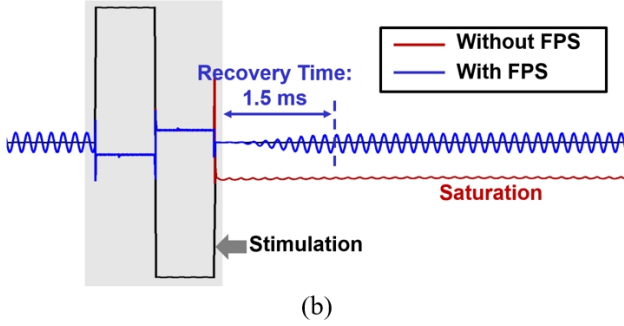
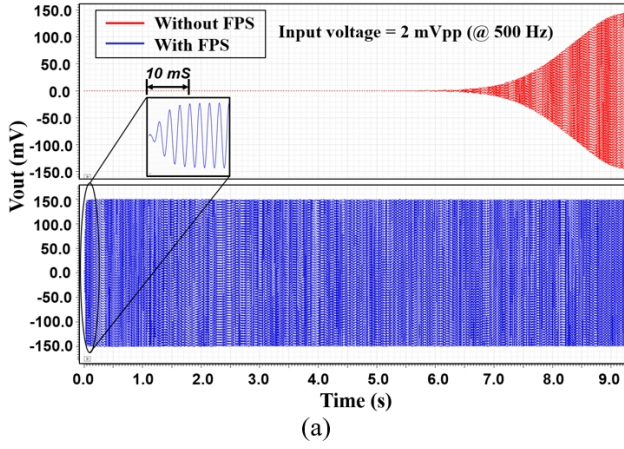


Fig. 7. Comparison of output response during (a) power-on and (b) stimulation with and without the floating power supply (FPS) module.

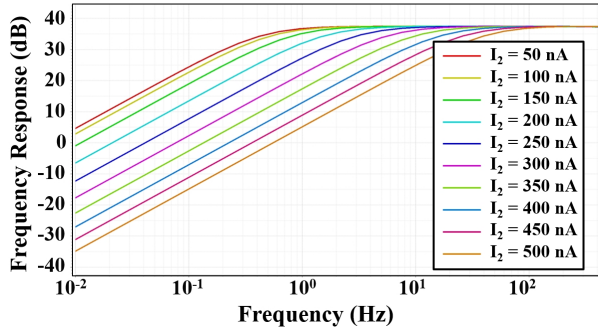


Fig. 8. High-pass corner frequency variation with different biasing currents I_2 .

To evaluate the settling time of the proposed A-PR, we conducted a comparison based on its output response with an amplifier utilizing T-PR in the feedback, activated by powering on the PMU module and providing stimulation at the input, as shown in Fig. 7. The only difference is that the T-PR does not employ a floating power supply (FPS). Results reveal that when the FPS is employed, the settling time is reduced to less than 10 ms, while the recovery time from stimulation is only 1.5 ms, nearly 1000 times shorter compared to the amplifier utilizing the T-PR.

The tunability of the circuit is depicted in Fig. 8. Instead of adjusting the biasing current I_0 of the floating voltage reference [8], we achieve tunability by modifying the value

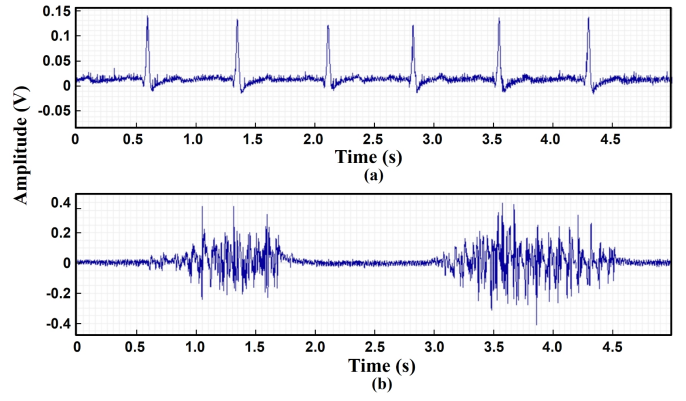


Fig. 9. Samples of data recorded from the human body: (a) Electrocardiograph (ECG) signal; (b) Electromyography (EMG) signal.

TABLE I
COMPARISON OF SETTLING TIME WITH PRIOR-ART PUBLICATIONS

	This work	TIM'22* [12]	ISCAS'18 [13]	TBioCAS'18 [14]	BioCAS'16 [15]
Input Referred Noise [μV_{rms}]	1.7 (0.5-3000Hz)	0.9 (0.5-1000Hz)	0.7 (0.5-100Hz)	2.02 (0.59-117Hz)	3.1 (0.5-1400Hz)
Settling Time	10 ms (Power on) 1.5 ms (Stimu.)	4.7 s (Stimu.)	30 ms** (Stimu.)	9 ms (Stimu.)	< 6.4 s (Stimu.)
THD [%]	0.45 @ 1mVpp, 200Hz	/	0.82 @ 100Hz	0.82 @ 70Hz	/
Self-Adaptive	Yes	Yes	No	Yes	Yes

* Built with commercial operational amplifiers. ** Analog part only.

of the pseudo-resistor (PR) through adjustments in the biasing current I_2 of the floating power supplies. This approach significantly simplifies the design of the biasing circuit, as I_0 is typically falls in the fA range, which can pose challenges for accurate implementation.

To validate the in vivo performance, the front-end system was interfaced with Ag/AgCl electrodes placed on a human subject. The recorded waveforms are depicted in Figure 9. Additionally, Table I offers a comparison with prior-art publications, demonstrating that our approach exhibits superior settling performance. Moreover, the self-adaptive nature of our chip obviates the need for additional control elements in the system.

IV. CONCLUSION

In this paper, we introduce self-adaptive pseudo-resistors (A-PRs) to tackle the long settling time associated with traditional pseudo-resistors in biomedical front-end circuits. By dynamically adjusting resistance based on the unsettled terminal voltage, A-PRs achieve fast settling times and excellent linearity without significant trade-offs in noise, power, or area. We validated our approach through simulations and on-chip measurements using a fabricated CCIA front-end in TSMC's 40-nm process technology, showing notable improvements in settling time, linearity, and tunability, making A-PRs a viable solution for biomedical applications.

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